Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 Calculate the potential and kinetic energy at the top and half-way down, and the energy transferred.

- Top: $E_p = 7.35$ J, $E_k = 0$ J
- Half-way: $E_p = 1.5 \times 9.8 \times 0.25 = 3.68$ J
- E_k half-way $= 7.35 - 3.68 = 3.68$ J
- 3.7 J has been transferred from potential to kinetic energy

2 A 3 kg buggy rolls from the top of an 80 cm ramp. Calculate the potential and kinetic energy at the top and half-way down.

- Top: $E_p = 3 \times 9.8 \times 0.8 = 23.5$ J, $E_k = 0$ J
- Half-way: $E_p = 3 \times 9.8 \times 0.4 = 11.8$ J
- E_k half-way $= 23.5 - 11.8 = 12$ J

3 A 5 kg mass is raised steadily through 2 m. Calculate the work done and the potential energy gained. What do you notice?

- $F = mg = 49$ N; $E_w = 49 \times 2 = 98$ J
- $E_p = 5 \times 9.8 \times 2 = 98$ J
- The work done lifting the mass equals the potential energy it gains.

4 A 20 kg mass falls freely. When it is 16 m above the ground its total energy is 3531 J. Calculate the height it started from.

- At the start it is at rest, so all 3531 J is potential energy
- $h = E_p \div (mg) = 3531 \div (20 \times 9.8)$
- $h = 18$ m

5 A 30 kg body is thrown straight up at 20 m/s. Calculate its kinetic energy and the height it rises to.

- $E_k = 0.5 \times 30 \times 20^2 = 6000$ J
- At the top $E_p = 6000$ J, so $h = 6000 \div (30 \times 9.8) = 20$ m

6 A 60 kg skateboarder starts at rest at the top of a 500 m hill. Ignoring friction, find the potential and kinetic energy at the top, the middle and the bottom.

- Top: $E_p = 60 \times 9.8 \times 500 = 294\,000$ J, $E_k = 0$
- Middle: $E_p = 147\,000$ J, $E_k = 147\,000$ J
- Bottom: $E_p = 0$, $E_k = 294\,000$ J